

MENTIS · Procurement Intelligence Brief

Substance: 0.9% saline

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Executive Summary

0.9% saline, or normal saline, is a widely utilized isotonic solution in fluid therapy for critically ill patients, serving essential roles in hydration, electrolyte management, and surgical care [source](#). The global market, valued at \$3.2 billion in 2023, is expected to grow at a 6% CAGR to reach \$5.8 billion by 2032 [source](#). Regulatory compliance focuses on good manufacturing practices and supply chain stability, including addressing prior shortages [source](#). Leading suppliers such as Baxter and Fresenius Kabi drive market innovation and capacity expansion, with North America dominating at 39.46% market share [source](#). Procurement risks include supply chain vulnerabilities and competition from balanced crystalloid alternatives, which offer potential clinical advantages but remain costlier [source](#).

Product Profile

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Clinical Use & Mechanism

0.9% saline, commonly referred to as "normal saline," is the most frequently utilized crystalloid solution in fluid therapy for critically ill patients. Its widespread application stems from its isotonic properties, making it a cornerstone for volume resuscitation and electrolyte management [source](#). Among crystalloid solutions, 0.9% saline is especially preferred for its affordability and accessibility compared to more complex fluid formulations, such as buffered solutions.

Despite its advantages, there is ongoing debate in clinical practice about the optimal type of intravenous fluid. Buffered solutions, for instance, may offer theoretical clinical advantages, including reduced risk of metabolic acidosis and fewer electrolyte disturbances. However, evidence comparing these solutions remains inconclusive, and usage varies significantly by institution and region [source](#).

The demand for 0.9% saline is driven by its versatility across a wide range of medical scenarios, including managing dehydration, hemorrhage, and perioperative care. As healthcare systems globally grapple with increasing patient volumes, the procurement and supply chain stability of 0.9% saline remain critical to ensuring effective clinical outcomes in acute and critical care settings. Policymakers and procurement teams must weigh its cost-efficiency against the potential benefits offered by alternative solutions in specific clinical situations.

Market Size & Demand Drivers

The 0.9% saline market continues to demonstrate robust growth, underpinned by its critical role in a wide range of medical applications and its designation as an essential medicine in 120 countries [source](#). The global market for normal saline was valued at USD 3.2 billion in 2023 and is projected to reach USD 5.8 billion by 2032, growing at a compound annual growth rate (CAGR) of approximately 6% [source](#).

Demand momentum is also reflected in broader estimates for the physiological saline market, which is expected to grow at a higher CAGR of 7.4%, increasing from USD 6.73 billion in 2025 to USD 9.99 billion by 2030, indicative of rising consumption across both established and emerging markets [source](#).

Key drivers include extensive use in hospitals and clinics, the largest end-use segment, where 0.9% saline is vital for intravenous infusions, wound care, and surgical procedures [source](#). Additionally, expanding surgical volumes, home care initiatives, and innovations such as preservative-free formulations are expected to sustain growth, particularly in developing regions [source](#).

However, market projections vary across sources, with some estimates suggesting slightly lower demand potential by 2035 in line with the pharmaceutical-grade sodium chloride segment's projected growth trajectory [source](#). Such discrepancies highlight uncertainties around demand modeling and scope definitions.

Top Manufacturers & Suppliers

The 0.9% saline market is characterized by contributions from several global manufacturers with diversified operations and market strategies. Key players include Baxter International Inc., Fresenius Kabi AG, B. Braun Melsungen AG, ICU Medical Inc., and Pfizer Inc. These companies dominate market share through extensive product portfolios, which encompass innovations such as prefilled syringes and preservative-free options [source](#). Baxter, for example, received FDA approval for its facility in Sabinanigo, Spain, enabling it to supply 0.9% saline in various volumes to the U.S. market [source](#).

Regionally, North America leads with a 39.46% market share as of 2025, driven by robust infrastructure, key manufacturers' presence, and initiatives to expand capacity during periods of shortage [source](#). Simultaneously, the physiological saline market shows significant activity across countries such as India, the Netherlands, and the United States, among others, reflecting its high global demand [source](#).

Additionally, strategic partnerships play a role in distribution. For instance, Grifols partnered with Henry Schein to distribute saline solutions in the U.S. market, leveraging Henry Schein's vast network [source](#). These dynamics indicate a highly competitive and interconnected market with emphasis on capacity, innovation, and strategic collaborations.

Regulatory & Compliance

The regulatory oversight surrounding 0.9% saline, commonly referred to as sodium chloride solution, focuses on manufacturing practices, storage conditions, and responses to supply chain disruptions.

According to the FDA, the production facilities for sodium chloride 0.9% Freeflex Injection Solution are inspected for compliance with good manufacturing practices. Notably, the Fresenius Kabi Norge AS plant in Halden, Norway, passed its most recent FDA inspection without objection, affirming that the importation and distribution of these products meet regulatory standards [FDA facility compliance](#).

Storage requirements for 0.9% saline solutions are governed by guidelines ensuring product stability. Manufacturers have established protocols to store the product at controlled room temperatures between 20–25°C (68–77°F), avoiding freezing conditions as per USP guidelines [USP storage protocol](#). Additionally, the stability of commercial batches is monitored throughout the assigned shelf life, with annual reporting ensuring continuous compliance with FDA requirements [stability monitoring](#).

Recent regulatory concerns include addressing the U.S. shortage of IV saline solutions, which the FDA announced had been resolved. During the shortage, demand for 0.9% saline faced significant strain, underlining the critical role of supply chain reliability in healthcare settings [FDA shortage resolution](#). Contrasting these advancements, a recall by Baxter International highlighted operational risks. Two lots of IV saline were voluntarily recalled due to particulate contamination from insects found through customer complaints [IV saline recall](#).

These dynamics illustrate the complexity of maintaining regulatory compliance for critical healthcare products in fluctuating market conditions.

Sourcing & Pricing Channels

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- **Sodium Chloride 0.9%, 1000mL Bag - Bound Tree Medical** — [Tavily](#) | false not grp product
Sodium Chloride 0.9%, 1000mL Bag |

| Product # | 601324X | | Volume | 1000mL | | Refrigeration | Refrigeration Not Required | | NDC | 0338-0049-04 | | Concentration | 0.9% | | Type | Bag |

Related Products

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Sodium Chloride Injection 0.9% IV Solution,

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Henry Schein 0.9% Sodium Chloride Injection, USP (10 mL bottle) sh

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- **Sodium Chloride 0.9% (Normal Saline), USP, Sterile Grade, Cytiva ...** — [Tavily](#) Fisher Scientific - part of Thermo Fisher Scientific

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Risks & Alternatives

0.9% saline, commonly referred to as normal saline, is a widely used intravenous fluid composed of 9 grams of sodium chloride per liter of water, which exceeds the salt concentration of blood [source](#). Despite its ubiquity in North America, particularly in emergency and perioperative settings, the solution is associated with potential risks, such as metabolic acidosis, acute kidney injury, and increased mortality rates [source](#).

Emerging alternatives, including balanced crystalloid solutions such as lactated Ringer's and Plasma-Lyte, offer advantages by incorporating electrolytes like potassium and calcium to better mimic the plasma composition. Research from two large trials, SALT-ED and SMART, underscores the marginal but notable benefits of balanced solutions over saline, particularly in reducing complications like hyperchloremic acidosis [source](#).

Recent challenges to normal saline's dominance include supply chain interruptions. For instance, prefilled saline flush syringes were flagged as a shortage item by the FDA in March 2022, demonstrating the vulnerability of supply channels [source](#). Additionally, product recalls, such as those involving Nurse Assist LLC in November 2023, due to contamination concerns, highlight risks related to specific manufacturers [source](#).

Moving forward, procurement strategies should weigh the clinical benefits of balanced solutions, while monitoring regulatory updates and addressing supply chain vulnerabilities proactively.

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